IIT BOMBAY TECHFEST 2026–27 | NAMMASPACE CHALLENGE

Standard Operating Procedure (SOP): Monocular Spatial Capture

Standardized Protocol for High-Fidelity Smartphone 3D Gaussian Splatting (20% Scoring Weightage)

1. Pre-Capture Environment Preparation

1. Static Space Enforcement: Moving individuals, pets, and rotating fans must be minimized during recording. In dynamic environments, median photometric loss during optimization down-weights transient pedestrians.

2. Illumination Consistency: Turn on all overhead artificial lights to provide even, diffuse lighting across the entire venue. Close blinds to eliminate moving sun glare. Never switch lights on or off during capture.

3. Featureless Surface Mitigation: On featureless blank white walls, attach temporary non-destructive markers (sticky notes or room numbers) to provide visual anchor points for COLMAP feature matching.

4. Specular Surfaces: Cover full-length mirrors and reflective glass panes or record at oblique angles to avoid virtual inverted reflections that corrupt depth estimation.

2. Hardware & Camera Configuration

Parameter	Specification	Engineering Rationale
Device	Standard Smartphone (iOS / Android)	Monocular setup without LiDAR (receives higher scoring priority).
Resolution	4K UHD (3840×2160) preferred; min 1080p	Higher resolution provides denser pixel gradient feedback during optimization.
Frame Rate	60 FPS (or 30 FPS)	High frame rate reduces motion blur between adjacent temporal samples.
Lens	Primary 1x Wide Lens (24–28mm equivalent)	Avoids 0.5x ultra-wide optical barrel distortion.
Exposure/Focus	**LOCKED (AE/AF Lock)**	Prevents dynamic brightness shifts that corrupt photogrammetric color consistency.
Shutter Speed	≥ 1/120 s	Eliminates motion blur; motion blur turns directly into scattered needle splats.

3. The Triple-Loop Trajectory (Capture Pattern)

To guarantee the mandatory ≥75% frame overlap, capture follows a continuous 3-loop nested trajectory:

1. **Loop 1 — Outer Perimeter (Facing Walls, 0° Pitch):** Walk slowly around the outer perimeter at eye height (~1.6 m) facing walls, doors, and signage. Maintain steady heel-to-toe walking.
2. **Loop 2 — Inner Incline (Facing Center, 30° Pitch Down):** Walk the perimeter at chest level (~1.0 m) angled downward toward the floor, desks, and baseboards to anchor ground plane geometry.
3. **Loop 3 — Central Serpentine & 360° POI Orbits:** Step into the center, walk primary corridors, and execute continuous 360° circular orbits around key landmarks (meeting tables, lounge furniture, entrance

doors).

4. **Mandatory Loop Closure:** The recording must finish at the exact physical coordinate where it started, allowing bundle adjustment in COLMAP to bind camera drift to <0.67 px and eliminate ground tilt.

4. Reconstruction & Floater Pruning Pipeline

- **Frame Extraction:** Extract frames at **7 to 10 FPS** via FFmpeg with scene-filtering to eliminate motion-blurred frames (`ffmpeg -i input.mp4 -vf "fps=7,mpdecimate" frames/%05d.png`).
- **Pose Estimation:** Headless COLMAP using the `OPENCV` camera model with exhaustive feature matching.
- **SOTA 3DGS Optimization:** Train using **2D Gaussian Splatting (2DGS)** or **Nerfstudio MCMC** for 30,000 steps on GPU infrastructure, enforcing planar surface reconstruction on walls, floors, and ceilings.
- **Floater Elimination:** Prune primitives with opacity $\alpha < 0.05$, prune splats with radius > 0.5 m, and apply a 3D bounding box spatial crop.
- **WebGL Export:** Export as compressed `.splat` or quantized 16-bit `.ksplat` for instant 60 FPS streaming in the NammaSpace viewer.